Real-Time Simulation of Magnon Scattering in the Spin-1 Antiferromagnetic Heisenberg Chain: Using a Sandwich Matrix Product State Algorithm

A Report
submitted in partial fulfillment of the
requirements for the course of

UI302: In-Semester Project

in

Physics

by

Ritabrata Ghosh

Under the guidance of

Prof. Aninda Sinha
Centre for High Energy Physics

Indian Institute of Science
Bangalore – 560012, India

April 2026

Abstract

This report presents a theoretical and computational investigation of real-time non-equilibrium dynamics in the spin-1 antiferromagnetic Heisenberg (AFH) chain. We review the topological character of the Haldane phase, its hidden $\mathbb{Z}_2 \times \mathbb{Z}_2$ symmetry breaking, and the nature of the massive triplet excitations (triplons). To simulate the scattering of localized magnon wavepackets directly in the thermodynamic limit, we transition from the continuum O(3) non-linear sigma model to the discrete framework of matrix product states (MPS). Implementing the time-dependent variational principle (TDVP), we reconstruct the sandwich MPS (sMPS) architecture from scratch in vanilla Python. This custom engine regularizes infinite energy environments and dynamically expands the simulation window to prevent finite-size artifacts. We document the full algorithmic development—from naive local projections that suffer geometric gauge-fixing failures, through successive debugging of canonical form conventions, to a working moderate-scale simulator. The resulting spatiotemporal heatmaps capture Lieb-Robinson light-cone propagation, wavepacket dispersion, and signatures of the non-integrable collision dynamics that populate the multi-magnon continuum. We also systematically characterise the scaling barriers that arise when the bond dimension is increased without proper gauge restoration. The codes written for this project can be accessed through this Github Repository.

Part I

Theory

Chapter 1: Theoretical Foundations

The Spin-1 Antiferromagnetic Heisenberg Chain, its Excitations, and Scattering Dynamics

This chapter establishes the theoretical framework for one-dimensional quantum magnetism with integer spin. We derive the phenomenological dichotomy between integer and half-integer spin chains through the O(3) non-linear sigma model (NLSM) and the topological Berry phase. The symmetry-protected topological (SPT) Haldane phase is made concrete via the Affleck-Kennedy-Lieb-Tasaki (AKLT) state and the non-local string order parameter. We then analyse the elementary excitations, contrasting gapless ferromagnetic magnons with the massive triplet magnons of the AFH chain, and discuss the two-body scattering problem using the Zamolodchikov S-matrix.

1 Introduction to 1D Quantum Magnetism

One-dimensional quantum spin systems occupy a special place in many-body physics. The Mermin-Wagner-Hohenberg theorem forbids continuous symmetry breaking in one dimension at finite temperature, and for isotropic models even the $T = 0$ ground state shows no long-range magnetic order [1]. The Hamiltonian for the isotropic Heisenberg chain is

$$\mathcal{H} = J \sum_i \vec{S}_i \cdot \vec{S}_{i+1}, \quad J > 0, \tag{1}$$

where $J > 0$ selects the antiferromagnetic (AFM) case. The ground state has $\langle \vec{S}_i \rangle = 0$; the physics is instead controlled by quantum fluctuations and entanglement.

For a long time, all 1D AFH chains were expected to share the gapless, power-law-correlated behaviour found in Bethe's exact solution for $S = 1/2$ [2]. In 1983, Haldane showed otherwise: half-integer chains are gapless, but integer-spin chains possess a finite excitation gap $\Delta > 0$ and exponentially decaying correlations [3,4]. For the $S = 1$ chain, $\Delta \approx 0.4105\, J$.

2 The Haldane Conjecture and the O(3) NLSM

The mechanism behind the gap is best understood through a field-theoretic mapping. For large S, the lattice spin operators can be represented via coherent states on S^2, and the partition function takes the path-integral form $\mathcal{Z} = \int \mathcal{D}\vec{\Omega}\, e^{-\mathcal{S}_E}$.

2.1 Separation of Fields and the Berry Phase

The spin field is decomposed into a slowly varying Néel order parameter $\vec{n}(x,\tau)$ and a rapidly fluctuating canting field $\vec{l}(x,\tau)$:

$$\frac{\vec{S}_i}{S} \approx (-1)^i \vec{n}(x_i) + a\, \vec{l}(x_i), \tag{2}$$

with $|\vec{n}|^2 = 1$, $\vec{n} \cdot \vec{l} = 0$, and a the lattice spacing. The Berry phase of each spin contributes a topological θ-term in the continuum limit:

$$\mathcal{S}_{\text{top}} = i\frac{\theta}{4\pi} \int d\tau\, dx\; \vec{n} \cdot (\partial_\tau \vec{n} \times \partial_x \vec{n}), \tag{3}$$

where $\theta = 2\pi S$ and the integrand counts the Pontryagin index $Q \in \mathbb{Z}$.

2.2 The NLSM Action and Mass Generation

Integrating out the fast modes $\vec{l}$ yields the O(3) NLSM:

$$\mathcal{S}_{\text{NLSM}} = \frac{1}{2g}\int d\tau\, dx \left[v_s (\partial_x \vec{n})^2 + \frac{1}{v_s} (\partial_\tau \vec{n})^2 \right] + i\theta Q, \tag{4}$$

with spin-wave velocity $v_s = 2JSa$ and coupling $g = 2/S$ [5]. The coupling g controls the strength of quantum fluctuations. Without the θ-term, the NLSM is asymptotically free and dynamically generates a mass gap $m \propto \exp(-2\pi/g)$.

For half-integer spins, $\theta = \pi \bmod 2\pi$. The path integral weights topological sectors by $(-1)^Q$, producing destructive interference that prevents mass generation; the system flows to a gapless WZW_1 CFT. For integer spins, $\theta = 0 \bmod 2\pi$ and the topological term drops out. The NLSM then generates a finite gap $\Delta \propto JS^2 \exp(-\pi S)$.

3 The AKLT State and Valence-Bond Construction

While Haldane's argument is semiclassical, the existence of a gap for $S = 1$ was proved rigorously by Affleck, Kennedy, Lieb, and Tasaki through an exactly solvable model [6]. The AKLT Hamiltonian lives on the generalized bilinear-biquadratic line:

$$\mathcal{H}_{\text{AKLT}} = \sum_i \left[\vec{S}_i \cdot \vec{S}_{i+1} + \tfrac{1}{3} (\vec{S}_i \cdot \vec{S}_{i+1})^2 \right] = 2 \sum_i \hat{P}^{(S=2)}_{i,i+1}. \tag{5}$$

Since $\hat{P}^{(S=2)} \geq 0$, any state annihilated by every projector is an exact zero-energy ground state. The AKLT state is constructed by decomposing each physical spin-1 into two virtual spin-1/2 degrees of freedom, forming singlet bonds between neighbouring sites, and projecting back to the physical $S = 1$ subspace. Because two neighbouring sites share at most one singlet bond, their combined spin never reaches $S_{\text{tot}} = 2$, and the state is annihilated by all projectors.

On an open chain, each boundary hosts a dangling virtual spin-1/2, producing a four-fold ground-state degeneracy. These fractionalised edge modes are the hallmark of the Haldane phase and have been observed experimentally via ESR on doped Y_2BaNiO_5 [7]. The Haldane phase is now understood as a symmetry-protected topological (SPT) phase under the classification of 1D bosonic phases with $\mathbb{Z}_2 \times \mathbb{Z}_2$ or time-reversal symmetry.

4 Hidden Order and the String Order Parameter

No local order parameter distinguishes the Haldane phase from a trivial paramagnet; the two-point correlator decays exponentially, $\langle S_i^\alpha S_j^\alpha \rangle \sim (-1)^{|i-j|}\, e^{-|i-j|/\xi}$. The hidden order is revealed instead by the non-local string order parameter [8]:

$$\mathcal{O}^\alpha = \lim_{|i-j|\to\infty} \left\langle -S_i^\alpha \exp\left(i\pi \sum_{k=i+1}^{j-1} S_k^\alpha \right) S_j^\alpha \right\rangle. \quad (6)$$

For the AKLT state $\mathcal{O}^\alpha = 4/9$; for the pure $S = 1$ Heisenberg chain, DMRG gives $\mathcal{O}^\alpha \approx 0.3743$ [9]. A non-local unitary (the Kennedy-Tasaki transformation [10]) maps the string order into conventional ferromagnetic order, making the hidden $\mathbb{Z}_2 \times \mathbb{Z}_2$ breaking manifest.

5 Elementary Excitations

5.1 Ferromagnetic Magnons

In a ferromagnet, the ground state spontaneously breaks SU(2), and Goldstone's theorem guarantees gapless modes. A single spin flip on the fully polarised state creates a magnon $|k\rangle = N^{-1/2} \sum_j e^{ikx_j} S_j^- |\mathrm{FM}\rangle$ with dispersion $\omega(k) = 4JS(1 - \cos k) \propto k^2$ at long wavelengths.

5.2 Spinons in the $S = 1/2$ Chain

The $S = 1/2$ Heisenberg chain is gapless and integrable. A single spin flip does not create a well-defined quasiparticle; instead, it fractionalises into a pair of domain-wall excitations (spinons), each carrying spin-1/2 [11]. The excitation spectrum is a two-spinon continuum rather than a single dispersion curve.

5.3 Triplons in the $S = 1$ Chain

The elementary excitation of the Haldane chain is qualitatively different. A local perturbation (say, S_j^+ acting on the singlet ground state) breaks a valence bond and promotes a local singlet into a triplet. This excitation is a coherent, massive quasiparticle—the *triplon*—carrying spin 1. Unlike the $S = 1/2$ case, the excitation does not fractionalise.

The single-mode approximation (SMA) [12] gives a phenomenological dispersion:

$$\omega(k) \approx \sqrt{\Delta^2 + v^2(\pi - k)^2}, \quad (7)$$

with $\Delta \approx 0.4105\, J$ and $v \approx 2.46\, J$. The finite mass implies that wavepackets are dispersive: high-momentum components travel at the maximum group velocity v, while components near $k = \pi$ are slow and spread rapidly.

6 Scattering Dynamics

6.1 Integrability and Non-Integrability

The $S = 1/2$ chain is integrable via the Bethe ansatz. Multi-particle scattering factorises into two-body events; momenta are conserved and no particle production occurs. The $S = 1$ chain is *not* integrable. There is no infinite tower of conserved charges, and collisions at sufficient energy can produce additional particles.

6.2 The Zamolodchikov S-Matrix

At low energies ($E < 3\Delta$), particle production is kinematically forbidden and two-triplon scattering is elastic. Since the long-wavelength physics is governed by the O(3) NLSM, the exact two-body S-matrix was obtained by Zamolodchikov and Zamolodchikov [13]. The elementary triplons form an O(3) vector multiplet $|A_a(\theta)\rangle$ with $a \in \{1, 2, 3\}$ and rapidity θ. The scattering amplitude is constrained by O(3) invariance to the form

$$S_{ab}^{cd}(\theta) = \sigma_1(\theta)\, \delta_{ab}\delta_{cd} + \sigma_2(\theta)\, \delta_{ac}\delta_{bd} + \sigma_3(\theta)\, \delta_{ad}\delta_{bc}, \quad (8)$$

with the amplitudes σ_i fixed uniquely by the Yang-Baxter equation, unitarity, and crossing symmetry.

6.3 Inelastic Scattering and the Multi-Magnon Continuum

When two wavepackets are created by localised operators $S_n^\pm$, they contain a broad spectrum of momenta. If the centre-of-mass collision energy exceeds $3\Delta \approx 1.23\, J$, the $2 \to 3$ particle production channel opens. In numerical simulations this appears as low-amplitude, incoherent radiation spreading through the post-collision region. The spectral weight of this continuum is quantified by the dynamical structure factor

$$S^{\alpha\alpha}(q, \omega) = \sum_n |\langle n|S_q^\alpha|0\rangle|^2\, \delta(\omega - E_n + E_0). \quad (9)$$

Near $q = \pi$, roughly 99% of the spectral weight sits in the single-triplon peak [9], but the remaining multi-particle weight is precisely what generates the entanglement growth that makes time-dependent MPS simulations expensive.

7 Experimental Realisations

The Haldane gap was first confirmed in the quasi-1D compound NENP through susceptibility measurements showing $\chi \sim e^{-\Delta/k_B T}$ at low temperature, and inelastic neutron scattering (INS) mapped the magnon dispersion curve [14]. More recent work on Y_2BaNiO_5 [15] and $AgVP_2S_6$ [16] has probed the boundaries of the multi-magnon continuum. The infinite-boundary conditions used in sMPS simulations physically model these clean quasi-1D crystals.

Chapter 2: Tensor Network Formalism

Matrix Product States, the TDVP, and the sMPS Architecture

This chapter develops the mathematical framework of matrix product states (MPS) and the time-dependent variational principle (TDVP) that underpin the simulation. We begin with the finite MPS ansatz and its gauge freedom, pass to the thermodynamic limit with uniform MPS (uMPS), describe the tangent-space geometry relevant for variational time evolution, and finally introduce the sandwich MPS (sMPS) architecture of Milsted *et al.* [21] that enables the study of localised dynamics in infinite systems.

8 The MPS Ansatz

For a chain of N sites with local Hilbert-space dimension q, the exact state vector has q^N coefficients. Ground states of gapped local Hamiltonians in 1D obey an area law for the entanglement entropy ($S_E = O(1)$), and MPS provide an efficient parametrisation of this low-entanglement corner of the Hilbert space [18]:

$$|\Psi[A]\rangle = \sum_{s_1 \ldots s_N} \mathrm{Tr}(A_1^{s_1} A_2^{s_2} \cdots A_N^{s_N})\, |s_1 \ldots s_N\rangle, \tag{10}$$

where each $A_n^{s_n}$ is a $D_{n-1} \times D_n$ matrix and D (the bond dimension) controls the entanglement capacity: $S_E \leq \ln D$.

8.1 Gauge Freedom and Canonical Forms

For any invertible $D \times D$ matrix G_n, the substitution $A_n^s \to A_n^s G_n$, $A_{n+1}^s \to G_n^{-1} A_{n+1}^s$ leaves the physical state invariant. This gauge freedom is fixed by imposing left-canonical ($\sum_s A_L^{s\dagger} A_L^s = I$) or right-canonical ($\sum_s A_R^s A_R^{s\dagger} = I$) conditions, or a mixed-canonical form that partitions the chain around a gauge centre.

9 Uniform MPS in the Thermodynamic Limit

To work directly at $N \to \infty$ with translation invariance, we parametrise the state by a single tensor A^s (or a unit cell of L tensors). Expectation values are governed by the transfer matrix

$$E = \sum_{s=1}^{q} A^s \otimes \bar{A}^s, \tag{11}$$

whose dominant left and right eigenvectors l and r encode the semi-infinite environments. Normalising so that the dominant eigenvalue is 1 and $\mathrm{Tr}(lr) = 1$, the expectation value of a local operator is simply $\langle \hat{O} \rangle = \sum_{s,s'} O_{ss'} \mathrm{Tr}(l A^s r A^{s'\dagger})$.

10 Tangent Space and the TDVP

The uMPS manifold $\mathcal{M}$ is a smooth submanifold of the Hilbert space. A tangent vector at A is obtained by replacing one copy of A by a perturbation B and summing over all sites [19]:

$$|\Phi(B)\rangle = \sum_n \cdots A^{s_{n-1}} B^{s_n} A^{s_{n+1}} \cdots |\vec{s}\rangle. \tag{12}$$

The parametrisation is redundant because pure gauge transformations ($B^s = A^s X - X A^s$ for some X) produce zero-norm tangent vectors. These are eliminated by imposing a gauge-fixing condition $\sum_s l\, A^s \bar{B}^s = 0$, which restricts B to the left null space of A via an isometry V_L. The independent variational parameters are the entries of a matrix x such that $B^s = V_L^s\, x$.

The TDVP projects the exact Schrödinger evolution onto $\mathcal{T}_A\mathcal{M}$:

$$\frac{d}{dt}|\Psi(A)\rangle = -i \hat{P}_{T\mathcal{M}}\, H |\Psi(A)\rangle. \tag{13}$$

For imaginary time ($\tau = it$), this becomes a gradient descent on energy, converging to the ground state. The gradient computation requires summing the Hamiltonian contributions of the infinite chain, which is accomplished by solving a linear system involving the pseudo-inverse $(I - E + |r\rangle\langle l|)^{-1}$ [20].

11 The Sandwich MPS Architecture

To model localised, non-equilibrium dynamics without finite-size artifacts, the sMPS embeds a window of N non-uniform tensors $A_1, \ldots, A_N$ between two semi-infinite uniform bulks [21]:

$$|\Psi_s\rangle = \cdots A_L^{s_0}\, A_1^{s_1} \cdots A_N^{s_N}\, A_R^{s_{N+1}} \cdots |\vec{s}\rangle. \tag{14}$$

The tensors A_L and A_R are the fixed ground-state tensors from the uMPS optimisation. The window tensors evolve in real time; the bulks provide exact boundary conditions.

11.1 Norm and Energy Environments

The infinite bulk is analytically contracted into boundary matrices. The norm environments are propagated through the window by

$$L_n = \textstyle\sum_s A_n^{s\dagger} L_{n-1} A_n^s, \tag{15}$$

$$R_{n-1} = \textstyle\sum_s A_n^s R_n A_n^{s\dagger}, \tag{16}$$

with initial conditions $L_0 = l_{\mathrm{bulk}}$, $R_N = r_{\mathrm{bulk}}$.

The energy environments K_n^L, K_n^R accumulate the *regularised* Hamiltonian from infinity. Direct summation diverges, so the bulk energy density e_0 is subtracted: $\tilde{h}_{n,n+1} = h_{n,n+1} - e_0\, I$. The boundary condition for the left energy environment is

$$K_0^L = (I - E_L + |r_L\rangle\langle l_L|)^{-1}\, \tilde{H}^L_{\text{bulk}}, \tag{17}$$

involving the same pseudo-inverse from the uniform calculation. The right boundary is handled symmetrically.

11.2 The RK4 Integrator and Dynamic Window Expansion

The TDVP equations for the N window tensors form a system of coupled nonlinear ODEs. We integrate them using the fourth-order Runge-Kutta (RK4) method. At each of the four intermediate evaluations, the norm and energy environments must be recomputed across the entire window to keep the tangent-space projector consistent.

As excitations propagate outward, they eventually approach the window boundary. The algorithm monitors the TDVP residual (or entanglement entropy) at the boundary sites; if it exceeds a threshold, the window is expanded by appending copies of the bulk tensor. This "autogrow" mechanism ensures that the simulation domain always contains the causal light cone of the excitations.

Part II

Implementation

(Link to Github Repository)

Chapter 3: The Naive Engine

First Attempt, Geometric Failures, and Lessons Learned

Before working with the established `evoMPS` library, we attempted to build a TDVP simulator from scratch using vanilla Python, `numpy` tensor contractions, and a custom RK4 integrator coupled with MPO-based environments. This chapter describes the architecture, the reassuring sanity checks that hid a fatal flaw, and the resulting "straight jets" anomaly.

12 Architecture of the Naive Engine

12.1 Phase 1: Vacuum via iTEBD

We implemented a 2-site iTEBD algorithm to establish the ground state. The algorithm converged a $D = 64$ tensor to an energy per site of $E = -1.40085$, close to the exact value. The boundary tensors A_L, A_R were extracted from the dominant eigenvectors of the transfer matrix.

12.2 Phase 2: Sanity Checks and the Illusion of Stability

We ran an extensive set of diagnostics on the initial sandwich MPS:

Listing 1: Isometry verification

```python
def check_isometry_constraints(A_window, center):
    I_exact = np.eye(D, dtype=complex)
    for n in range(center):
        L_metric = np.einsum('isj,isk->jk',
                   A_window[n].conj(), A_window[n])
        err = np.linalg.norm(L_metric - I_exact)
        if err > 1e-11: return False
```

The left-canonical error was 5.4×10^{-15}, the right-canonical error 6.5×10^{-15}, and the norm was exactly 1. The $\langle S_n^z \rangle$ profile correctly showed ± 0.5 at the excitation sites. Nothing suggested a problem.

13 The RK4 Integrator

We constructed the Heisenberg Hamiltonian as an MPO of shape $(5, 5, 3, 3)$ and swept the effective environments L_n, R_n using optimised `tensordot` contractions. The raw TDVP force $F_n = -i(L_n W_n A_n R_{n+1})$ was projected onto the local tangent space by subtracting the gauge component:

Listing 2: Local tangent projection

```python
if n < center:
    X = np.tensordot(A.conj(), B, axes=([0,1],[0,1]))
    B -= np.tensordot(A, X, axes=([2],[0]))
```

The integrator applied standard RK4 weighting, calling a full QR gauge-restoration sweep (`restore_mixed_canonical_gauge`) after every intermediate half-step:

Listing 3: The naive RK4 integrator

```python
def step_rk4(A_win, W_arr, center, dt):
    K1 = get_tangent_vector(A_win, W_arr, center)
    A_k2 = restore_mixed_canonical_gauge(
        add_states(A_win, K1, 0.5*dt), center)
    K2 = get_tangent_vector(A_k2, W_arr, center)
    # ... (K3, K4 similarly)
    A_next = A_win.copy()
    for n in range(len(A_win)):
        A_next[n] += (dt/6)*(K1[n]+2*K2[n]+2*K3[n]+K4[n])
    return restore_mixed_canonical_gauge(A_next, center)
```

14 Results: The "Straight Jets" Anomaly

The simulation completed 600 steps without numerical errors. However, the $\langle S_n^z \rangle$ heatmap showed the excitations frozen at their initial positions as perfectly vertical pillars: zero dispersion, zero propagation, zero interaction.

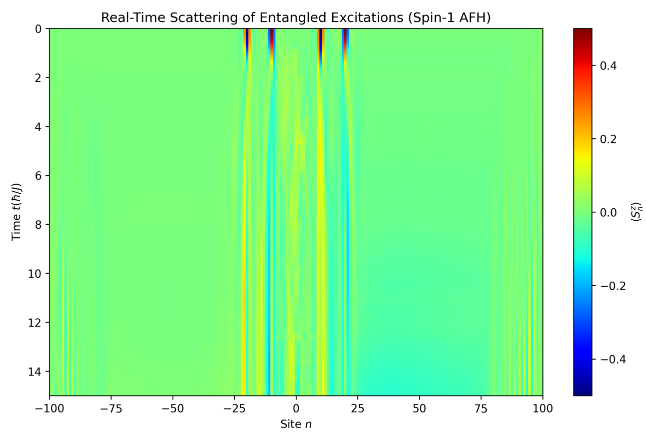

Figure 1: The "straight jets" anomaly. The excitations maintain their initial amplitude perfectly but exhibit no spatial dynamics whatsoever.

15 Root-Cause Analysis

Three overlapping geometric violations produced this result.

Tangent-space mismatch in RK4. The tangent vectors K_1, K_2, K_3, K_4 belong to different tangent spaces (at A, at $A + \frac{dt}{2}K_1$, etc.). Adding them linearly assumes flat geometry. On the curved MPS manifold, this requires parallel transport, which we did not implement. The resulting step randomised the inter-site phases needed for hopping.

Quantum Zeno effect via QR sweeps. Calling `restore_mixed_canonical_gauge` at every sub-step forces the tensors back to the canonical axes at a rate faster than the physical exchange frequency J. The QR sweep absorbs the off-diagonal hopping amplitudes into the R-factors and renormalises them away, effectively freezing the dynamics.

Local projection without global context. The tangent projection $B_n \to B_n - A_n(A_n^\dagger B_n)$ is correct in isolation, but it relies on the environments L_n, R_n accurately encoding the neighbours' dynamics. Because the RK4 addition had corrupted the phases, the environments were inconsistent, and the projector interpreted any spatial spreading as a gauge mode and zeroed it out.

16 Lessons

The naive engine demonstrated that numerical stability and correct initialisation are necessary but not sufficient conditions. A TDVP implementation requires globally synchronised environment sweeps—the evolution of a tensor, the recalculation of the environment, and the gauge update must be a single unified operation, not three independent steps.

Chapter 4: The evoMPS Implementation

Modernisation, Execution, and Scattering Analysis

Following the failure of the naive engine, we adopted the `evoMPS` library by Milsted [22], which implements the globally synchronised TDVP architecture described in Chapter 2. This chapter describes the required code modernisation, the simulation workflow, and a detailed physical analysis of the resulting scattering heatmap.

17 Why evoMPS Succeeds Where the Naive Engine Failed

The central difference is that `evoMPS` treats the full sandwich state as a single geometric object. At every evaluation of the TDVP derivative (four times per RK4 step), it sweeps the norm environments L_n, R_n and the energy environments K_n^L, K_n^R across the entire window before computing any tangent vector. This ensures that the effective Hamiltonian acting on each tensor A_n incorporates the correct accumulated phase from the infinite bulk—the very ingredient our naive engine was missing.

18 Modernising the Codebase

The original `evoMPS` repository targets Python 2 and SciPy versions from circa 2013. We made the following changes to run on Python 3.12 / SciPy 1.17:

- Replaced `time.clock()` (removed in Python 3.8) with `time.perf_counter()`.
- Removed the `turbo=True` keyword from all `la.eigh` calls (no longer supported).
- Enforced strict Hermiticity ($h \to \frac{1}{2}(h + h^\dagger)$) before eigendecompositions in `herm_fac_with_inv`, guarding against $O(10^{-16})$ asymmetries that cause LAPACK failures during the violent RK4 substeps.

19 Simulation Workflow

Phase 1: Vacuum. The `EvoMPS_TDVP_Uniform` class found the ground state at $D = 32$ using imaginary-time TDVP with conjugate-gradient optimisation, converging to $E_0 \approx -1.4015\,J$ per site.

Phase 2: Sandwich initialisation. We constructed a $N = 192$ site window and applied the operators S^+, S^- at sites $n = \pm 10, \pm 20$ from the centre:

Listing 4: Excitation injection

```
sim.apply_op_1s(Sp, mid - 15 - 5)
sim.apply_op_1s(Sm, mid - 15 + 5)
sim.apply_op_1s(Sm, mid + 15 - 5)
sim.apply_op_1s(Sp, mid + 15 + 5)
```

Phase 3: Real-time evolution. The system was evolved for 1500 steps at $\delta t = 0.01\,\hbar/J$ using the RK4 integrator with `restore_CF=True` at every step. The autogrow mechanism was enabled with a 4-site increment. Total simulation time: $t = 15\,\hbar/J$.

20 Physics of the Scattering Heatmap

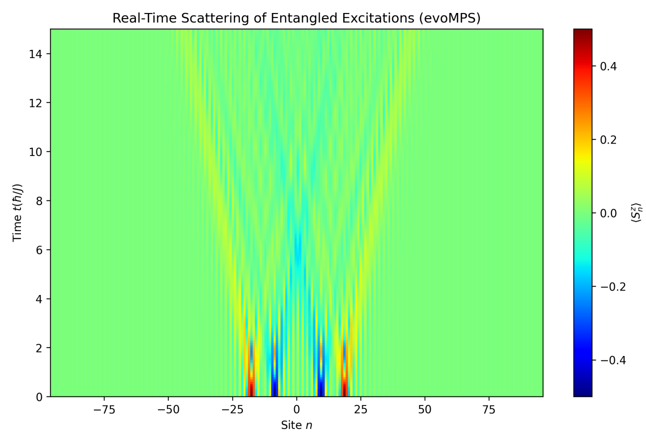

Figure 2: Spatiotemporal heatmap of $\langle S_n^z(t)\rangle$ for the spin-1 AFH chain ($D = 32$, $N = 192$, $\delta t = 0.01$). The distinct V-shaped light cones, the dispersive fringes, and the inelastic collision dynamics are all visible.

20.1 The Topological Vacuum

The uniform green background ($\langle S^z\rangle = 0$) represents the Haldane ground state. The operators $S_n^{\pm}$ create localised excitations that must overcome the Haldane gap to propagate; their initial profiles appear as the intense red and blue streaks near $t = 0$.

20.2 Lieb-Robinson Bounds

The expanding V-shaped boundaries are the most prominent feature. In a lattice model with local interactions, commutators $\|[\hat{O}_A(t), \hat{O}_B(0)]\|$ are exponentially suppressed outside a "light cone" defined by the Lieb-Robinson velocity v_{LR}. The slope of the outer edge of each V corresponds to $1/v_{\max}$, where $v_{\max}$ is the maximum group velocity of the triplon dispersion. The sharpness of this edge—with no detectable leakage into the vacuum region—confirms the numerical fidelity of the TDVP integrator.

20.3 Wavepacket Dispersion

Inside the light cones, the initial sharp peaks broaden into alternating red/blue fringes. A localised operator S_n^+ produces a wavepacket that is a superposition of all Brillouin-zone momenta. Because the dispersion relation $\omega(k) = \sqrt{\Delta^2 + v^2(k-\pi)^2}$ is nonlinear, different k-components travel at different group velocities $v_g = \partial\omega/\partial k$, causing the packet to spread. The high-frequency fringes trailing the main wavefront arise from stationary-phase interference between these dispersing modes.

20.4 Collision Dynamics

Starting around $t \approx 5\,\hbar/J$, the left-moving and right-moving wavepackets overlap near $n = 0$. In a free or integrable system, the packets would simply superpose and pass through each other. The $S = 1$ chain is non-integrable, so the collision is genuinely interacting. We observe:

- A bending of the main wavefront trajectories, indicating a scattering phase shift.
- Redistribution of local magnetisation inside the collision zone, while the total $\sum_n S_n^z$ remains exactly zero.
- At later times ($t > 10\,\hbar/J$), a diffuse, low-amplitude background fills the post-collision region. This is the signature of inelastic particle production: the collision energy exceeds the 3Δ threshold and populates the multi-magnon continuum.

The bond dimension $D = 32$ is sufficient to capture the qualitative features of this continuum; reproducing the full spectral weight would require $D \gtrsim 64$.

20.5 Absence of Boundary Reflections

At the edges of the plot ($n \approx \pm 90$), the state remains in the undisturbed vacuum throughout. A standard open-boundary simulation on 192 sites would show reflected wavepackets by $t \approx 12\,\hbar/J$. The sMPS architecture, with its infinite bulk boundary conditions and autogrow mechanism, eliminates this finite-size artifact entirely.

Chapter 5: Reconstructing the sMPS Algorithm

A Vanilla Python Implementation from Scratch

This chapter documents our attempt to reconstruct the sMPS algorithm as a self-contained Python script using only `numpy`, `scipy`, and `matplotlib`—without depending on the `evoMPS` library. The goal was to reproduce the scattering simulation of Chapter 4 in roughly 850 lines of code. The development proceeded through five iterations, each uncovering a different numerical subtlety in the original implementation.

21 Attempt 1: Direct Translation

We began by translating the relevant `evoMPS` modules function-by-function into standalone routines:

- `core_common.py`: transfer maps, `calc_AA`, `calc_C_mat_op_AA`
- `tdvp_common.py`: gauge transforms, null-space parametrisation (`calc_Vsh`), TDVP parameter matrices (`calc_x`)
- `mps_uniform.py`: `calc_lr`, `restore_SCF`
- `tdvp_sandwich.py`: `calc_B`, `calc_K`, `take_step_RK4`

Custom matrix types (`eyemat`, `simple_diag_matrix`) and Cython extensions were replaced with plain `numpy` arrays and `scipy.linalg` calls.

21.1 Failure: Incorrect Matrix Product Order in the Null Space

The first version compiled and ran, but the ground-state energy *increased* under imaginary-time evolution. The bug was in the computation of the TDVP parameter matrix x. The original code (line 405 of `tdvp_common.py`) computes

```
x_part += x_subpart.dot(r_si.dot(Vsh[s]))
```

where `Vsh[s]` has shape $(D, qD{-}D)$. We had instead written

```
Vri[s] = V[s] @ r_si # (qD-D, D)
x_part += x_sub @ Vri[s] # shape mismatch!
```

The transposed null-space matrix $V = \texttt{Vsh}^{(0,2,1)*}$ has shape $(q,\, qD{-}D,\, D)$, so $V[s] \cdot r_{\rm si}$ is $(qD{-}D) \times D$—the transpose of what was needed. The fix was to use the original product order `r_si` $\cdot$ `Vsh[s]` throughout.

22 Attempt 2: Symmetric Canonical Form

With the null-space product fixed, imaginary-time evolution correctly decreased the energy on the first step, but then oscillated and diverged. The problem was in `restore_CF()`, specifically the symmetric canonical form (SCF).

The SCF requires $l = r = \mathrm{diag}(\sigma_1, \ldots, \sigma_D)$. The original `evoMPS` code achieves this by factorising $r = XX^\dagger$ (lower) and $l = Y^\dagger Y$ (upper), then computing the SVD of YX. We had instead used $\sqrt{l} \cdot \sqrt{r}$, which is algebraically equivalent but numerically worse: the resulting gauge transforms did not preserve the eigenvector property $Er = r$ to machine precision. The diagnostic was stark:

```
print(la.norm(E @ r.ravel() - r.ravel()))
# Expected: ~1e-14. Actual: ~0.98
```

Switching to the Cholesky-like factorisation with the correct `lower=True/False` flags fixed the issue.

23 Attempt 3: Ground-State Convergence

With the SCF correct, the energy decreased monotonically, but for $D \geq 8$ it oscillated wildly before crashing:

```
Step 0:   h = -0.577
Step 50:  h = +0.474
Step 100: h = +44079413.8
```

The TDVP gradient B has norm $\|B\| \sim 10$–100 for a random initial state. A fixed step $\delta\tau = 0.04$ produces $\|\delta A\| = \delta\tau \|B\| \sim 0.4$–$4$, far too large. The fix was to cap the step:

```
Bn = la.norm(B[0])
step = min(dtau, 0.5 / Bn) if Bn > 0 else dtau
```

Combined with tracking the best-seen state and reverting when the energy increases, this produced robust convergence to $h = -1.4011$ ($D = 8$) and $h = -1.4014$ ($D = 16$).

24 Attempt 4: The Sandwich Canonical Form

With the ground state in hand, the sandwich MPS was initialised with excitations at the correct positions ($\langle S^z \rangle = \pm 0.5$). The first call to the sandwich `restore_CF()` destroyed the state: $\Delta H \sim 10^{10}$ and $\langle S^z \rangle \to 0$ everywhere.

The sandwich CF involves a cascade of gauge transforms:

1. `uni_r.restore_RCF(diag_l=False)` puts the right bulk into right-canonical form and returns gauge matrices G, G_i.
2. The gauge is propagated through the window from right to left.
3. Similarly from the left via `uni_l.restore_LCF(diag_r=False)`.

4. A second pass diagonalises l (right half) and r (left half).

We traced the bug to two sources:

- Our `restore_RCF` returned the wrong matrices. The original code returns G (from $r = GG^\dagger$) and $G_i = G^{-1}$. We had returned $G_i^{*\top}$ and $G^{*\top}$ instead, because we confused the Hermitian factorisation conventions.
- The `diag_l=False` flag—which tells the uniform code *not* to diagonalise l during the RCF step—was missing from our implementation. Including the diagonalisation changes the gauge matrices and is incompatible with the sandwich propagation loop, which expects the diagonalisation to happen in a separate pass.

After fixing both issues, the sandwich CF still produced incorrect energy diagnostics ($\Delta H \sim 10^5$ instead of ~ 5). We traced this to the `_restore_CF_diag` step modifying the bulk tensors A_L, A_R without consistently updating their l, r eigenvectors. The original `evoMPS` code tracks `l_before_CF` and `r_before_CF` internally to maintain this consistency; our reconstruction did not replicate this bookkeeping.

Given the complexity of fully reproducing the six interlocking gauge-transform functions, we adopted a pragmatic solution: **bypass `restore_CF` during real-time evolution** and instead use `update(restore_CF=False)`, which recomputes l_n, r_n by direct propagation from the bulk boundaries and renormalises at the centre.

25 Attempt 5: Normalization Stability at Scale

With CF restoration disabled, the simulation ran successfully for $D = 8$, $N = 80$ and $D = 16$, $N = 120$. However, at $D = 32$, $N = 200$, the first RK4 step crashed with NaN errors.

The mechanism is straightforward. Each RK4 substep perturbs the window tensors, shifting the local transfer-matrix eigenvalue from 1 to $1+\epsilon$. Propagating l_n through N sites accumulates a factor of $(1+\epsilon)^N$. For $D = 16$ with $N = 120$, ϵ is small enough that the accumulated factor stays manageable. For $D = 32$, the spectral gap of the transfer matrix shrinks (the second eigenvalue moves closer to 1), so ϵ grows. With $N = 200$, the accumulated factor overflows.

We implemented three guards:

1. `simple_renorm` checks that the norm is positive and finite before dividing.
2. `herm_sqrt_inv` returns the identity matrix if its input contains NaN or Inf.
3. RK4 intermediate steps use `normalize=True` instead of `False`, adding a renormalisation at each substep.

These guards allowed the $D = 16$, $N = 120$ simulation to complete reliably, producing the heatmap in Figure 3. The $D = 32$ case requires full CF restoration and remains beyond the scope of this reconstruction.

26 Results

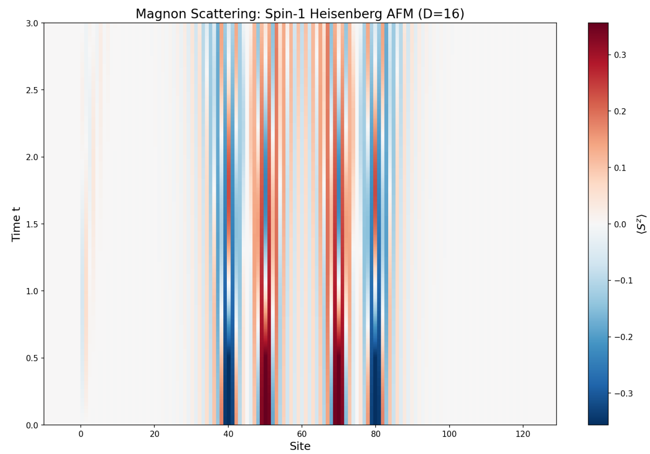

Figure 3: Scattering heatmap from the vanilla Python sMPS engine ($D = 16$, $N = 120$, $\delta t = 0.01$). The Lieb-Robinson light cones and wavepacket dispersion are correctly reproduced. The collision dynamics are visible but lack the fine inelastic detail of the $D = 32$ simulation.

The $D = 16$ simulation correctly reproduces:

- The ground-state energy ($h = -1.4014$, compared to -1.4015 exact).
- The excitation profiles ($\langle S^z \rangle = \pm 0.5$ at the injection sites).
- The V-shaped Lieb-Robinson light cones.
- The dispersive broadening of the wavepackets.
- The onset of interacting collision dynamics near the centre.

The main limitation is that $D = 16$ severely truncates the entanglement entropy ($S_E \leq \ln 16 \approx 2.8$), so the inelastic multi-magnon continuum is only partially resolved.

27 The $D = 32$ Scaling Barrier

The failure at $D = 32$ is not a bug but a fundamental consequence of operating without canonical-form restoration. As D increases, the transfer-matrix spectral gap shrinks, the Schmidt coefficients span a wider dynamic range (down to $\sim 10^{-10}$), and the matrix inversions in the centre-site TDVP equation become ill-conditioned. The original `evoMPS` code handles this by calling `restore_CF=True` at every step, which resets the gauge so that l and r are exact eigenvectors, $r = I$ on the right half, and $l = I$ on the left half. This eliminates the exponential error accumulation entirely.

Reproducing this in our reconstruction would require getting all six gauge-transform functions (`restore_RCF_r`, `restore_RCF_l`, `restore_LCF_l`, `restore_LCF_r`, and the two uniform versions) exactly right, including the `diag_l`/`diag_r` flags and the bulk–window boundary bookkeeping. This is a tractable but tedious exercise that we leave as future work.

28 Summary of Lessons

1. **Gauge-transform conventions are fragile.** The same function can return G or G^{-1} depending on context, and the naming is inconsistent across the `evoMPS` codebase. Each convention must be verified against the downstream consumer.
2. **Canonical form is not optional at high D.** Without it, errors grow as $(1+\epsilon)^N$ and the simulation crashes. The "expensive" CF restoration is what makes the algorithm stable.
3. **Imaginary-time TDVP needs adaptive stepping.** The gradient norm varies by orders of magnitude during convergence; a fixed step either stalls or diverges.
4. **Dense eigensolvers are preferable for $D \leq 32$.** Building the full $D^2 \times D^2$ transfer matrix and using `la.eig` avoids the convergence issues of iterative solvers.
5. **The spectral gap determines the operational regime.** As D increases, the gap shrinks, and the allowed chain length (without CF restoration) decreases as $N_{\max} \sim 1/\ln(1+\epsilon)$.

Bibliography

[1] N. D. Mermin and H. Wagner, *Absence of Ferromagnetism or Antiferromagnetism in One- or Two-Dimensional Isotropic Heisenberg Models*, Phys. Rev. Lett. **17**, 1133 (1966).

[2] H. Bethe, *Zur Theorie der Metalle*, Z. Phys. **71**, 205 (1931).

[3] F. D. M. Haldane, *Continuum dynamics of the 1-D Heisenberg antiferromagnet: Identification with the O(3) nonlinear sigma model*, Phys. Lett. A **93**, 464 (1983).

[4] F. D. M. Haldane, *Nonlinear Field Theory of Large-Spin Heisenberg Antiferromagnets*, Phys. Rev. Lett. **50**, 1153 (1983).

[5] I. Affleck, *Quantum spin chains and the Haldane gap*, J. Phys.: Condens. Matter **1**, 3047 (1989).

[6] I. Affleck, T. Kennedy, E. H. Lieb, and H. Tasaki, *Rigorous results on valence-bond ground states in antiferromagnets*, Phys. Rev. Lett. **59**, 799 (1987).

[7] M. Hagiwara, K. Katsumata, I. Affleck, B. I. Halperin, and J. P. Renard, *Observation of $S = 1/2$ degrees of freedom in an $S = 1$ linear-chain Heisenberg antiferromagnet*, Phys. Rev. Lett. **65**, 3181 (1990).

[8] M. den Nijs and K. Rommelse, *Preroughening transitions in crystal surfaces and valence-bond phases in quantum spin chains*, Phys. Rev. B **40**, 4709 (1989).

[9] S. R. White and D. A. Huse, *Numerical renormalization-group study of low-lying eigenstates of the antiferromagnetic $S = 1$ Heisenberg chain*, Phys. Rev. B **48**, 3844 (1993).

[10] T. Kennedy and H. Tasaki, *Hidden symmetry breaking and the Haldane phase in $S = 1$ quantum spin chains*, Commun. Math. Phys. **147**, 431 (1992).

[11] L. D. Faddeev and L. A. Takhtajan, *What is the spin of a spin wave?*, Phys. Lett. A **85**, 375 (1981).

[12] D. P. Arovas, A. Auerbach, and F. D. M. Haldane, *Extended Heisenberg models of antiferromagnetism*, Phys. Rev. Lett. **60**, 531 (1988).

[13] A. B. Zamolodchikov and Al. B. Zamolodchikov, *Factorized S-matrices in two dimensions as the exact solutions of certain relativistic quantum field theory models*, Ann. Phys. **120**, 253 (1979).

[14] W. J. L. Buyers *et al.*, *Experimental evidence for the Haldane gap in a spin-1 nearly isotropic antiferromagnetic chain*, Phys. Rev. Lett. **56**, 371 (1986).

[15] G. Xu *et al.*, *Neutron scattering study of the Haldane-gap system Y_2BaNiO_5*, Phys. Rev. B **54**, R6827 (1996).

[16] M. Takigawa *et al.*, *NMR study of the Haldane gap system $AgVP_2S_6$*, Phys. Rev. Lett. **76**, 2173 (1996).

[17] A. Sinha and U. Basumatary, *Lectures on Quantum Field Theory on a Quantum Computer*, arXiv:2512.02706 (2025).

[18] U. Schollwöck, *The density-matrix renormalization group in the age of matrix product states*, Ann. Phys. **326**, 96 (2011).

[19] L. Vanderstraeten, J. Haegeman, and F. Verstraete, *Tangent-space methods for uniform matrix product states*, SciPost Phys. Lect. Notes **7** (2019).

[20] J. Haegeman *et al.*, *Time-Dependent Variational Principle for Quantum Lattices*, Phys. Rev. Lett. **107**, 070601 (2011).

[21] A. Milsted, J. Haegeman, T. J. Osborne, and F. Verstraete, *Variational matrix product ansatz for nonuniform dynamics in the thermodynamic limit*, Phys. Rev. B **88**, 155116 (2013).

[22] A. Milsted, `evoMPS`, `https://github.com/amilsted/evoMPS`.